



# Math 4641: Numerical Methods

## Lecture 6 — Direct Method of Interpolation

**Syed Rifat Raiyan**

Lecturer

Department of Computer Science & Engineering  
Islamic University of Technology, Dhaka, Bangladesh

**Email:** rifatraiyan@iut-dhaka.edu

# Lecture Plan

## The agenda for today

- Understand the concept of interpolation
- Why polynomials are used as interpolating functions?
- Theorem about uniqueness of interpolating polynomials
- Find interpolant using the Direct method
- Use the interpolant to find the derivatives and integrals

# Interpolation

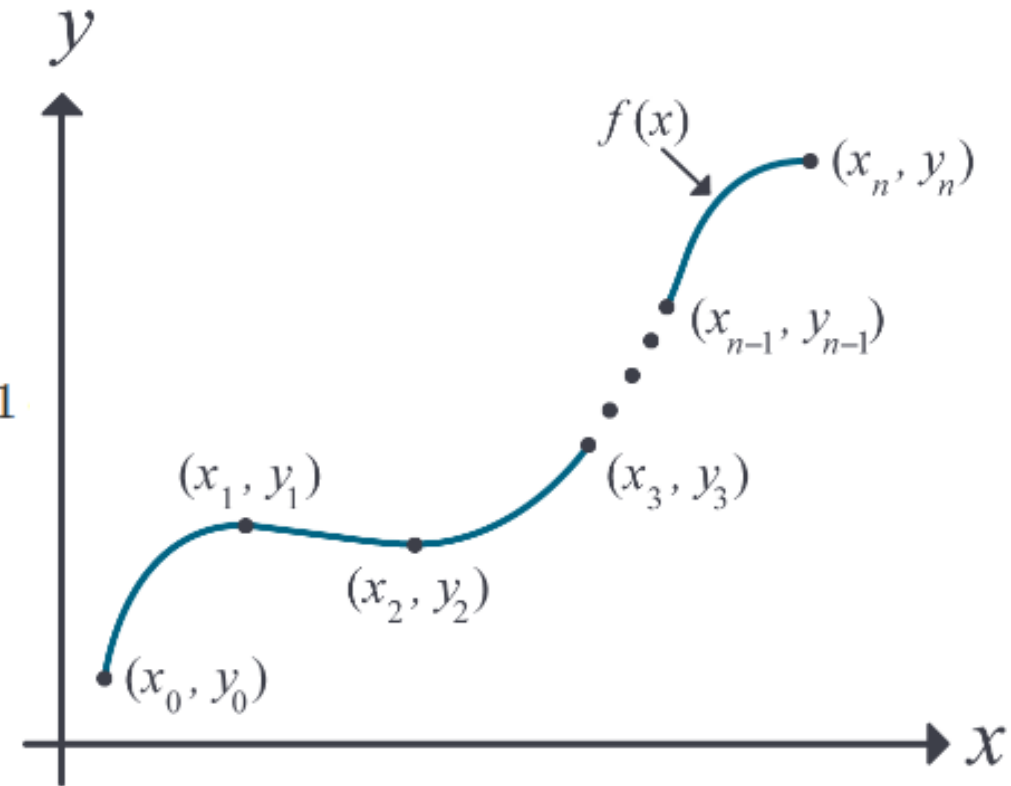
## What is it?

Interpolation is a type of estimation, a method of *constructing (finding) new data points* based on the range of a discrete set of *known* data points.

As given in Figure 1, data is given at discrete points such as  $(x_0, y_0), (x_1, y_1), \dots, (x_{n-1}, y_{n-1}), (x_n, y_n)$ .

A continuous function  $f(x)$  may be used to represent the  $n + 1$  data values with  $f(x)$  passing through the  $n + 1$  points.

Then one can find the value of  $y$  at any other value  $x$ .



**Figure 1.** Interpolation of a function given at discrete points

# Interpolation

## Why polynomial interpolants?

The function  $f(x)$  chosen for interpolation is called the *interpolant*.

A polynomial is a common choice for an interpolating function because polynomials are *easy* to —

- Evaluate
- Differentiate
- Integrate

relative to other choices such as a trigonometric and exponential series.

# Uniqueness of Polynomials

## Theorem

A polynomial of degree  $n$  or less that passes through  $n + 1$  data points is unique.

Let us use proof by contradiction. If the polynomial is not unique, then at least two polynomials of order  $n$  or less pass through the  $n + 1$  data points.

Assume two polynomials  $P_n(x)$  and  $Q_n(x)$  go through  $n + 1$  data points,

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$$

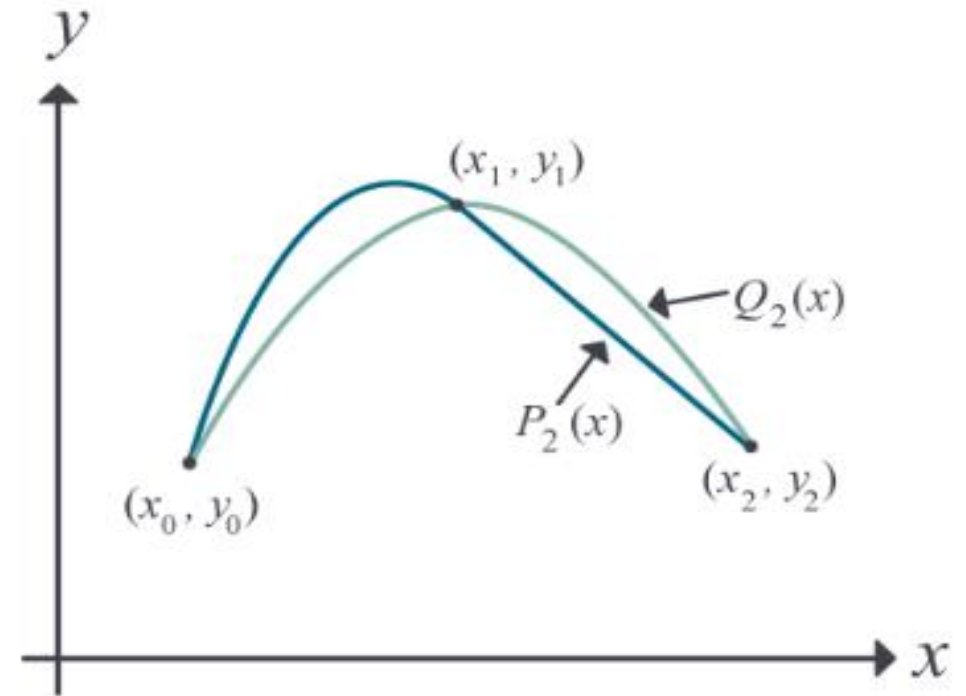
Then

$$R_n(x) = P_n(x) - Q_n(x) \quad (1)$$

Hence

$$R_n(x_i) = P_n(x_i) - Q_n(x_i) = 0, i = 0, \dots, n \quad (3)$$

The  $n^{\text{th}}$  order polynomial  $R_n(x)$  has  $n + 1$  zeros. A polynomial of order  $n$  can have  $n + 1$  zeros only if it is identical to a zero polynomial, that is,



$$R_n(x) \equiv 0 \quad (4)$$

Hence from Equation (1)

$$P_n(x) \equiv Q_n(x)$$

# Direct Method of Interpolation

## How does it work?

The direct method (also called the Vandermonde polynomial method) of interpolation is based on the following premise. Given  $n + 1$  data points, fit a polynomial of order  $n$  as given below

$$y = a_0 + a_1x + \dots + a_nx^n \quad (1)$$

through the data, where  $a_0, a_1, \dots, a_n$  are  $n + 1$  real constants. Since  $n + 1$  values of  $y$  are given at  $n + 1$  values of  $x$ , one can write  $n + 1$  equations. Then the  $n + 1$  constants,  $a_0, a_1, \dots, a_n$  can be found by solving the  $n + 1$  simultaneous linear equations. To find the value of  $y$  at a given value of  $x$ , simply substitute the value of  $x$  in Equation 1.

We *don't need* all the data points!

Instead we choose the *nearest* ones that *bracket* the unknown point.

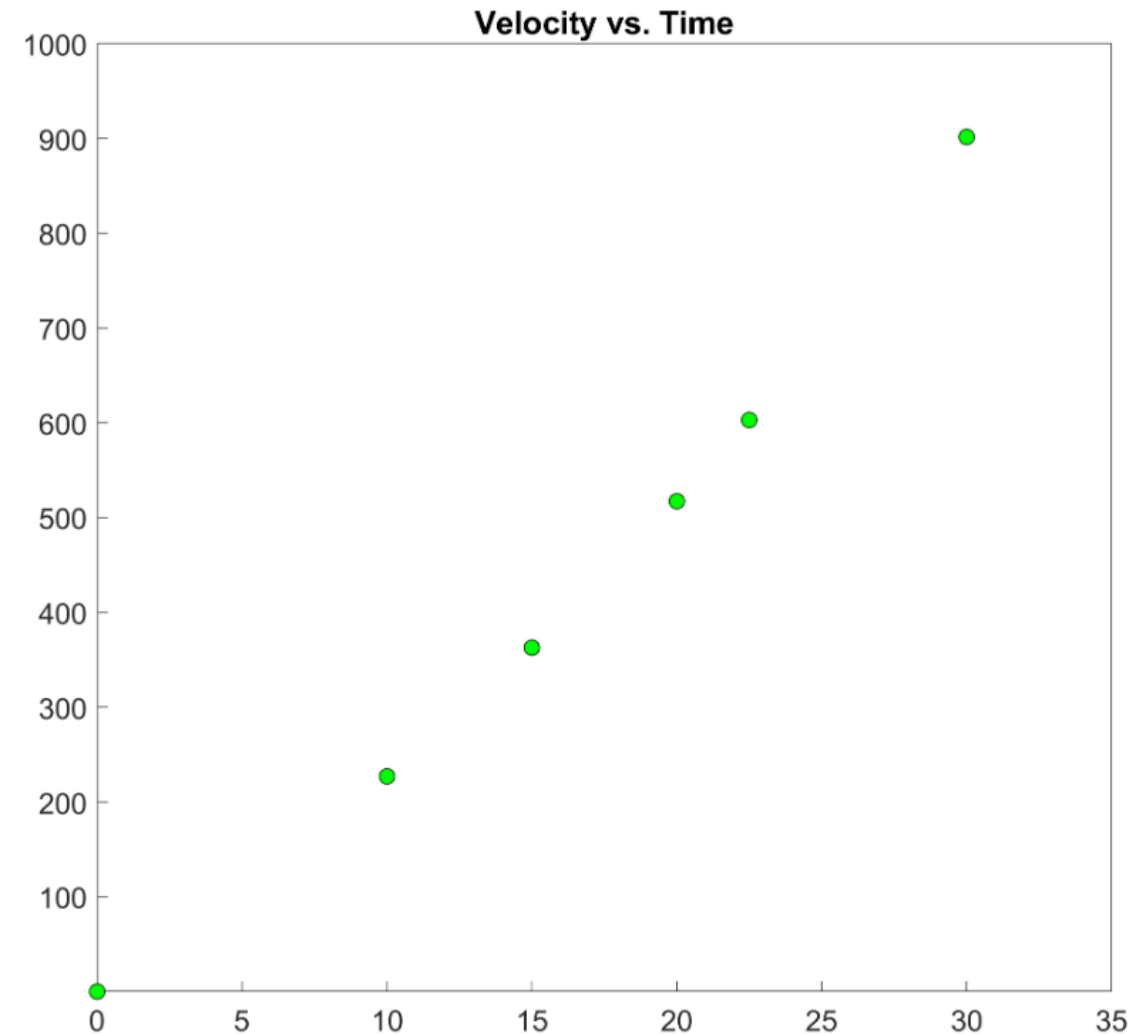
# Direct Method of Interpolation

## A first-order polynomial example

The upward velocity of a rocket is given as a function of time in Table 1.

**Table 1.** Velocity as a function of time.

| $t$ (s) | $v(t)$ (m/s) |
|---------|--------------|
| 0       | 0            |
| 10      | 227.04       |
| 15      | 362.78       |
| 20      | 517.35       |
| 22.5    | 602.97       |
| 30      | 901.67       |



**Figure 1.** Graph of velocity vs. time data for the rocket example.

Estimate the velocity at  $t = 16$  seconds using the direct method of interpolation with a first-order polynomial.

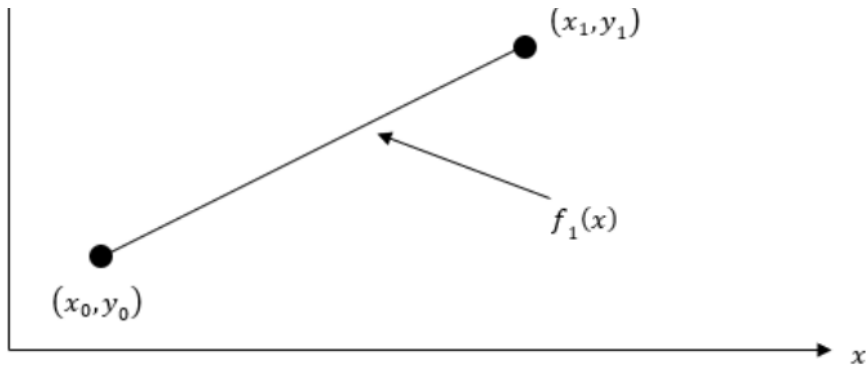
# Direct Method of Interpolation

## A first-order polynomial example

### Solution

For first-order polynomial interpolation (also called linear interpolation), the velocity given by

$$v(t) = a_0 + a_1 t \quad (E1.1)$$



**Figure 2.** Linear interpolation.

Since we want to find the velocity at  $t = 16$ , and we are using a first-order polynomial, we need to choose the two data points that are closest to  $t = 16$  that also bracket  $t = 16$  to evaluate it. The two points are  $t_0 = 15$  and  $t_1 = 20$ .

$$t_0 = 15, v(t_0) = 362.78$$

$$t_1 = 20, v(t_1) = 517.35$$

Equation (E1.1) gives

$$v(15) = a_0 + a_1(15) = 362.78$$

$$v(20) = a_0 + a_1(20) = 517.35$$

Writing the equations in matrix form, we have

$$\begin{bmatrix} 1 & 15 \\ 1 & 20 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 362.78 \\ 517.35 \end{bmatrix}$$

Solving the above two equations gives

$$a_0 = -100.93$$

$$a_1 = 30.914$$

Hence from Equation (E1.1)

$$\begin{aligned} v(t) &= a_0 + a_1 t \\ &= -100.93 + 30.914t, \quad 15 \leq t \leq 20 \end{aligned}$$

$$\begin{aligned} v(16) &= -100.92 + 30.914 \times 16 \\ &= 393.70 \text{ m/s} \end{aligned}$$



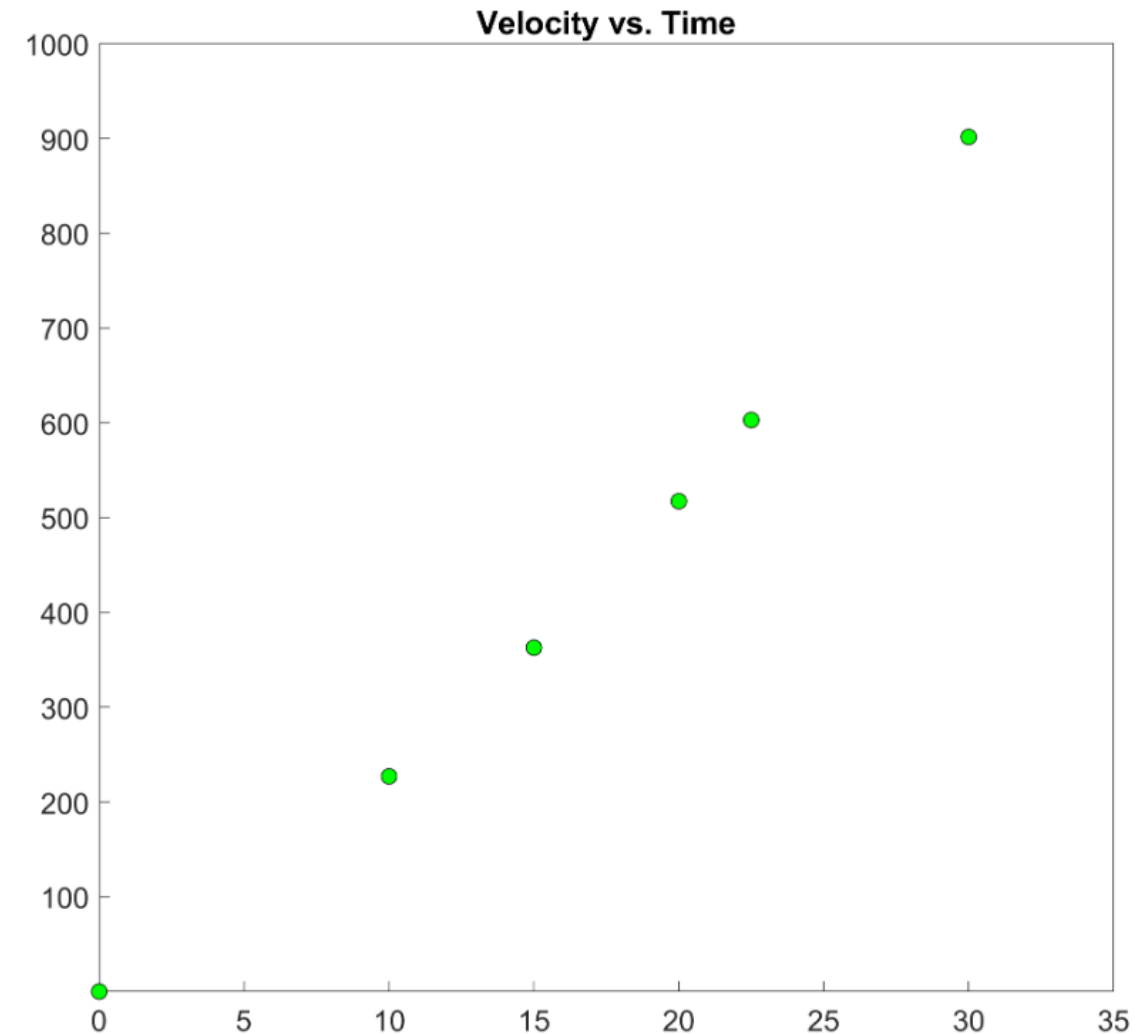
# Direct Method of Interpolation

## A second-order polynomial example

The upward velocity of a rocket is given as a function of time in Table 1.

**Table 1.** Velocity as a function of time.

| $t$ (s) | $v(t)$ (m/s) |
|---------|--------------|
| 0       | 0            |
| 10      | 227.04       |
| 15      | 362.78       |
| 20      | 517.35       |
| 22.5    | 602.97       |
| 30      | 901.67       |



**Figure 1.** Graph of velocity vs. time data for the rocket example.

- a) Estimate the velocity at  $t = 16$  seconds using the direct method of interpolation with a second-order polynomial.

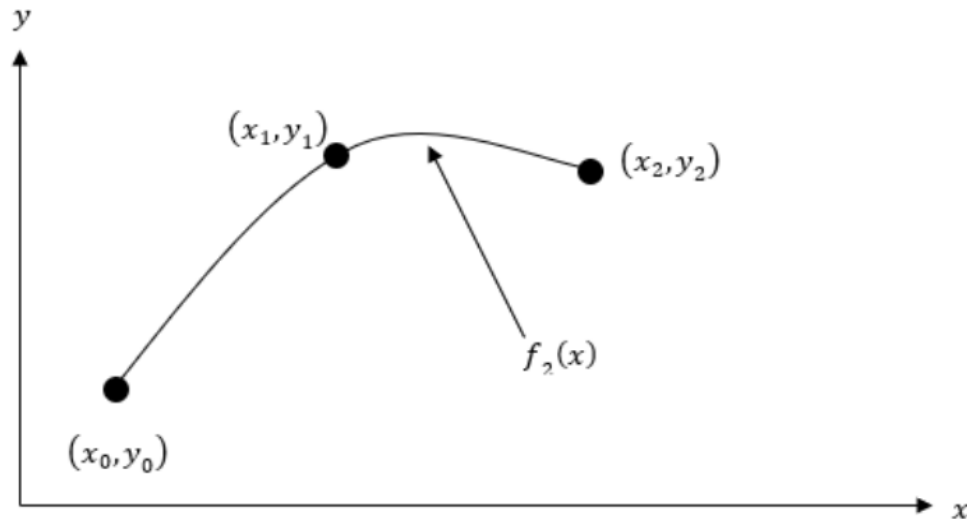
# Direct Method of Interpolation

## A second-order polynomial example

### Solution

For second-order polynomial interpolation (also called quadratic interpolation), the velocity is given by

$$v(t) = a_0 + a_1 t + a_2 t^2 \quad (E2.1)$$



**Figure 3.** Quadratic interpolation.

a) Since we want to find the velocity at  $t = 16$ , and we are using a second-order polynomial, we need to choose the three data points that are closest to  $t = 16$  that also bracket  $t = 16$  to evaluate it. The three points are  $t_0 = 10$ ,  $t_1 = 15$ , and  $t_2 = 20$ .

Then

$$t_0 = 10, v(t_0) = 227.04$$

$$t_1 = 15, v(t_1) = 362.78$$

$$t_2 = 20, v(t_2) = 517.35$$

Equation (E2.1) gives

$$v(10) = a_0 + a_1(10) + a_2(10)^2 = 227.04$$

$$v(15) = a_0 + a_1(15) + a_2(15)^2 = 362.78$$

$$v(20) = a_0 + a_1(20) + a_2(20)^2 = 517.35$$

# Direct Method of Interpolation

## A second-order polynomial example

Writing the three equations in matrix form, we have

$$\begin{bmatrix} 1 & 10 & 100 \\ 1 & 15 & 225 \\ 1 & 20 & 400 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 227.04 \\ 362.78 \\ 517.35 \end{bmatrix}$$

Solving the above three equations gives

$$a_0 = 12.050$$

$$a_1 = 17.733$$

$$a_2 = 0.37660$$

Hence from Equation (E2.1)

$$v(t) = 12.050 + 17.733t + 0.37660t^2, \quad 10 \leq t \leq 20$$

At  $t = 16$ ,

$$\begin{aligned} v(16) &= 12.050 + 17.7333(16) + 0.37660(16)^2 \\ &= 392.19 \text{ m/s} \end{aligned}$$

# Direct Method of Interpolation

## A second-order polynomial example

b) Find the absolute relative approximate error for the second-order polynomial approximation.

The absolute relative approximate error  $|\epsilon_a|$  obtained between the results from the first- and second-order polynomial is

$$\begin{aligned} |\epsilon_a| &= \left| \frac{392.19 - 393.70}{392.19} \right| \times 100 \\ &= 0.38410\% \end{aligned}$$

# Direct Method of Interpolation

## A second-order polynomial example

c) Using the second-order polynomial interpolant for velocity from part (a), find the distance covered by the rocket from  $t = 11$  s to  $t = 16$  s.

The distance covered by the rocket between  $t = 11$  s and  $t = 16$  s can be calculated from the interpolating polynomial (Equation E2.2)

$$v(t) = 12.050 + 17.733t + 0.37660t^2, \quad 10 \leq t \leq 20$$

Note that the polynomial is valid between  $t = 10$  s and  $t = 20$  s and hence includes the limits of integration of  $t = 11$  s and  $t = 16$  s can

So

$$\begin{aligned} s(16) - s(11) &= \int_{11}^{16} v(t) dt \\ &= \int_{11}^{16} (12.050 + 17.733t + 0.37660t^2) dt \\ &= \left[ 12.050t + 17.733\frac{t^2}{2} + 0.37660\frac{t^3}{3} \right]_{11}^{16} \\ &= 1604.3 \text{ m} \end{aligned}$$

# Direct Method of Interpolation

## A second-order polynomial example

d) Using the second-order polynomial interpolant for velocity from part (a), find the acceleration of the rocket at  $t = 16$  s.

The acceleration at  $t = 16$  s is given by

$$a(16) = \left. \frac{d}{dt} v(t) \right|_{t=16}$$

Given that from Equation (E2.2)

$$v(t) = 12.050 + 17.733t + 0.37660t^2, \quad 10 \leq t \leq 20$$

we get

$$\begin{aligned} a(t) &= \frac{d}{dt} v(t) \\ &= \frac{d}{dt} (12.050 + 17.733t + 0.37660t^2) \\ &= 17.733 + 0.75320t, \quad 10 \leq t \leq 20 \end{aligned}$$

Hence

$$\begin{aligned} a(16) &= 17.733 + 0.75320(16) \\ &= 29.784 \text{ m/s}^2 \end{aligned}$$